

1(a)	1 part enzyme extract/polyphenol oxidase with 9 parts of water / buffer ;	1
1(b)(i)	pH ;	1
1(b)(ii)	<i>any one from:</i> 1 temperature of, solutions / tubes ; 2 wavelength of, light / (light) filter (of 470 nm) ; 3 concentration of enzyme / polyphenol oxidase ; 4 total length of time /time reading interval ;	1
1(c)(i)	1 figures from printed line at or below 0.26 au and 30 s or figures from correct tangent ; 2 correct calculation / 0.0083 to 0.0089 / 8.3×10^{-3} to 8.9×10^{-3} ;	2
1(c)(ii)	sketched line below printed line starting at origin and constantly increasing to 180 s ;	1

Question	Answer	Marks
1(d)	1 state at least 5 dilutions (0.0–0.2%) of dopamine hydrochloride solution / substrate ; 2 describe a correct method to making at least 2 dilutions ; 3 add tubes to a water bath / incubator, at a, set / constant temperature (10–40 °C) ; 4 enzyme and substrate / (all) solutions equilibrated separately ; 5 state 2 different variables that are kept constant (volumes of enzyme and substrate or mixture / pH / filter or wavelength) ; 6 set to zero / calibrate, colorimeter using a, blank / water, at the start / between each reading ; 7 mix substrate and (buffered) enzyme and immediately add (colorimeter tube) to colorimeter ; or mix substrate and (buffered) enzyme and immediately start timing ; 8 measure absorbance at set time intervals (for set time) ; or record time at which red colour (first) appears ; 9 3 replicates / 2 repeats for same / one concentration and calculate a mean or 2 repeats, of experiment / each concentration, and calculate the means ; 10 calculate / find / work out rate (of reaction) ;	7

Question	Answer	Marks													
1(d)	11 named hazard and risk and precaution ; <table border="1"> <thead> <tr> <th>Hazard</th><th>Risk</th><th>Precaution</th></tr> </thead> <tbody> <tr> <td>dopamine hydrochloride</td><td>Toxic</td><td>don't drink don't pour down the sink wear gloves</td></tr> <tr> <td></td><td>Irritant / allergy</td><td rowspan="3">wear gloves / goggles / mask/ PPE</td></tr> <tr> <td>polyphenol oxidase / enzyme</td><td>Irritant / allergy</td></tr> <tr> <td>buffer</td><td>Irritant / allergy</td></tr> </tbody> </table> 12 use water/boiled enzyme instead of enzyme as the control ;	Hazard	Risk	Precaution	dopamine hydrochloride	Toxic	don't drink don't pour down the sink wear gloves		Irritant / allergy	wear gloves / goggles / mask/ PPE	polyphenol oxidase / enzyme	Irritant / allergy	buffer	Irritant / allergy	
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Question	Answer	Marks
1(e)	1 enzyme: from different (plant) sources / only from plants / enzyme not used from bananas ; 2 sources of information: maybe out of date / maybe unreliable / not comparable ; 3 variables: 2 variables not controlled / not standardised / no information (temperatures, pH, timings, volumes, concentrations) ; 4 stats: no statistical test / analysis ; 5 substrate: no information, on what substrate was used / whether tested on bananas or other fruits ; 6 inhibitor: do not know type of inhibition (competitive v. non-competitive) ; 7 anti-browning agents: <i>idea that</i> other anti-browning agents maybe more or less effective / may impact differently ; 8 anti-browning agents should be tested on all sources of enzyme / each only tested on one source of enzyme / may act differently on other sources of enzyme ; 9 validity: 2 independent variables ;	2
1(f)	sketched line (starts at origin) levels out showing lower Vmax ;	1P

Question	Answer	Marks
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